

Machine Learning A (2026–2027)

Home Assignment 3

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1 Optimal Classification (16 points)

Question 1

Define the set of possible y-classes $\mathcal{C}_k \in \{0, 1\}$. The optimal deterministic classification strategy is the one that maximizes the probability of success given x :

$$h(x) = \arg \max_{\mathcal{C}_k} \mathbb{P}(Y = \mathcal{C}_k \mid X = x) = \begin{cases} 0 & , \text{ if } x = 0 \\ 0 & , \text{ if } x = 1 \end{cases}.$$

The risk is calculated where we use the law of total probability and zero-one loss

$$\begin{aligned} \mathbb{E}[\ell(y, h(x))] &= \sum_{x \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y, X = x) \ell(y, h(x)) \\ &= \sum_{x \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(X = x) \mathbb{P}(Y = y \mid X = x) \ell(y, h(x)) \\ &= \mathbb{P}(X = 0) \mathbb{P}(Y = 0 \mid X = 0) \ell(0, h(0)) \\ &\quad + \mathbb{P}(X = 0) \mathbb{P}(Y = 1 \mid X = 0) \ell(1, h(0)) \\ &\quad + \mathbb{P}(X = 1) \mathbb{P}(Y = 0 \mid X = 1) \ell(0, h(1)) \\ &\quad + \mathbb{P}(X = 1) \mathbb{P}(Y = 1 \mid X = 1) \ell(1, h(1)) \\ &= \frac{1}{2} \cdot 1 \cdot 0 + \frac{1}{2} \cdot 0 \cdot 1 + \frac{1}{2} \cdot 0.8 \cdot 0 + \frac{1}{2} \cdot 0.2 \cdot 1 \\ &= \frac{1}{10}. \end{aligned}$$

Question 2

Using the fact that Y is conditionally independent of our hypothesis h' and that $h'(x) \equiv \mathbb{P}(h'(x) = \hat{y} \mid X = x) = \mathbb{P}(Y = \hat{y} \mid X = x)$ we calculate the risk of the hypothesis

$$\begin{aligned} \mathbb{E}[\ell(y, h'(x))] &= \sum_{\hat{y} \in \{0,1\}} \sum_{x \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y, X = x, h'(x) = \hat{y}) \ell(y, \hat{y}) \\ &= \sum_{\hat{y} \in \{0,1\}} \sum_{x \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y, h'(x) = \hat{y} \mid X = x) \mathbb{P}(X = x) \ell(y, \hat{y}) \\ &= \sum_{\hat{y} \in \{0,1\}} \sum_{x \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y \mid X = x) \mathbb{P}(h'(x) = \hat{y} \mid X = x) \mathbb{P}(X = x) \ell(y, \hat{y}) \\ &= \sum_{\hat{y} \in \{0,1\}} \sum_{x \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y \mid X = x) \mathbb{P}(Y = \hat{y} \mid X = x) \mathbb{P}(X = x) \ell(y, \hat{y}) \end{aligned}$$

$$\begin{aligned}
&= \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 1)\ell(1, 1) \\
&\quad + \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(X = 1)\ell(1, 0) \\
&\quad + \mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(X = 0)\ell(1, 1) \\
&\quad + \mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(X = 0)\ell(1, 0) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 1)\ell(0, 1) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(X = 1)\ell(0, 0) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(X = 0)\ell(0, 1) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(X = 0)\ell(0, 0) \\
&= \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(X = 1) \\
&\quad + \mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(X = 0) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 1) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(X = 0) \\
&= \frac{1}{5} \frac{4}{5} \frac{1}{2} + 0 \cdot 1 \cdot \frac{1}{2} + \frac{4}{5} \frac{1}{5} \frac{1}{2} + 1 \cdot 0 \cdot \frac{1}{2} \\
&= \frac{4}{25}.
\end{aligned}$$

So in this case the first hypothesis

$$h(x) = \arg \max_{\mathcal{C}_k} \mathbb{P}(Y = \mathcal{C}_k \mid X = x) = \begin{cases} 0 & , \text{ if } x = 0 \\ 0 & , \text{ if } x = 1 \end{cases},$$

is the optimal.

Question 3

Let the observed input be $X = x$. Then say $\mathbb{P}(h(x) = 1 \mid X = x) = \gamma_x$ and $\mathbb{P}(h(x) = 0 \mid X = x) = 1 - \gamma_x$ then the expected loss is

$$\begin{aligned}
\mathbb{E}[\ell(y, h(x)) \mid X = x] &= \sum_{\hat{y} \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y, h(x) = \hat{y} \mid X = x) \ell(y, \hat{y}) \\
&= \sum_{\hat{y} \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y \mid X = x) \mathbb{P}(h(x) = \hat{y} \mid X = x) \ell(y, \hat{y}) \\
&= \mathbb{P}(Y = 1 \mid X = x) \mathbb{P}(h(0) = 1 \mid X = x) \ell(1, 1) \\
&\quad + \mathbb{P}(Y = 1 \mid X = x) \mathbb{P}(h(0) = 0 \mid X = x) \ell(1, 0) \\
&\quad + \mathbb{P}(Y = 0 \mid X = x) \mathbb{P}(h(0) = 1 \mid X = x) \ell(0, 1) \\
&\quad + \mathbb{P}(Y = 0 \mid X = x) \mathbb{P}(h(0) = 0 \mid X = x) \ell(0, 0) \\
&= \mathbb{P}(Y = 1 \mid X = x) \mathbb{P}(h(0) = 0 \mid X = x) \\
&\quad + \mathbb{P}(Y = 0 \mid X = x) \mathbb{P}(h(0) = 1 \mid X = x) \\
&= \gamma_x \mathbb{P}(Y = 1 \mid X = x) + (1 - \gamma_x) \mathbb{P}(Y = 0 \mid X = x),
\end{aligned}$$

where the first equal sign uses the fact that Y is independent of our hypothesis h . Using the above we calculate the risk

$$\begin{aligned}
\mathbb{E}[\ell(y, h(x))] &= \mathbb{E}[\mathbb{E}[\ell(y, h(x)) \mid X = x]] \\
&= \sum_{x \in \{0,1\}} \mathbb{P}(X = x) \mathbb{E}[\ell(y, h(x)) \mid X = x] \\
&= \frac{1}{2} (\mathbb{E}[\ell(y, h(x)) \mid X = 0]) + \frac{1}{2} (\mathbb{E}[\ell(y, h(x)) \mid X = 1]) \\
&= \frac{1}{2} (\gamma_0 \mathbb{P}(Y = 1 \mid X = 0) + (1 - \gamma_0) \mathbb{P}(Y = 0 \mid X = 0)) \\
&\quad + \frac{1}{2} (\gamma_1 \mathbb{P}(Y = 1 \mid X = 1) + (1 - \gamma_1) \mathbb{P}(Y = 0 \mid X = 1))
\end{aligned}$$

Because this is linear in the probabilities the answer is immediately apparent. When minimizing w.r.t. γ_0 and γ_1 the optimal strategy will deterministically choose the output that is most likely given $X = x$, in most cases. If the probabilities of $Y = 0$ and $Y = 1$ are equal then the weights γ_x does not matter. Formally:

$$\gamma_x^* = \mathbb{P}(h(x) = 1 \mid X = x) = \begin{cases} 1 & , \text{ if } \mathbb{P}(Y = 1 \mid X = x) > \mathbb{P}(Y = 0 \mid X = x) \\ 0 & , \text{ if } \mathbb{P}(Y = 1 \mid X = x) < \mathbb{P}(Y = 0 \mid X = x) \\ [0, 1] & , \text{ if } \mathbb{P}(Y = 1 \mid X = x) = \mathbb{P}(Y = 0 \mid X = x) \end{cases}$$

2 Logistic Regression (50 points)

2.1 Cross-entropy error measure (16 points)

2.1.1 Rewriting negative logarithmic likelihood as cross-entropy

Firstly I rewrite $-\ln [\mathbb{P}(Y = y_n \mid X = \mathbf{x}_n; \mathbf{w})]$ where I use the fact that the sample space of the outcome variable is $Y \in \{-1, 1\}$.

$$\begin{aligned} -\ln [\mathbb{P}(Y = y_n \mid X = \mathbf{x}_n; \mathbf{w})] &= \mathbb{I}_{(y_n=1)} \{-\ln [\mathbb{P}(Y = 1 \mid X = \mathbf{x}_n; \mathbf{w})]\} \\ &\quad + \mathbb{I}_{(y_n=-1)} \{-\ln [\mathbb{P}(Y = -1 \mid X = \mathbf{x}_n; \mathbf{w})]\} \\ &= \mathbb{I}_{(y_n=1)} \{-\ln [\mathbb{P}(Y = 1 \mid X = \mathbf{x}_n; \mathbf{w})]\} \\ &\quad + \mathbb{I}_{(y_n=-1)} \{-\ln [1 - \mathbb{P}(Y = 1 \mid X = \mathbf{x}_n; \mathbf{w})]\} \\ &= \mathbb{I}_{(y_n=1)} \{-\ln [h(\mathbf{x}_n)]\} + \mathbb{I}_{(y_n=-1)} \{-\ln [1 - h(\mathbf{x}_n)]\} \\ &= \mathbb{I}_{(y_n=1)} \ln \left[\frac{1}{h(\mathbf{x}_n)} \right] + \mathbb{I}_{(y_n=-1)} \ln \left[\frac{1}{1 - h(\mathbf{x}_n)} \right] \end{aligned}$$

where $\mathbb{I}_{(\cdot)}$ is the indicator function. Inserting this into the general negative logarithmic likelihood function yields the wanted result:

$$\begin{aligned} \mathcal{L}(\mathbf{w}) &= -\frac{1}{N} \sum_{n=1}^N \ln [\mathbb{P}(Y = y_n \mid X = \mathbf{x}_n; \mathbf{w})] \\ &= \frac{1}{N} \sum_{n=1}^N -\ln [\mathbb{P}(Y = y_n \mid X = \mathbf{x}_n; \mathbf{w})] \\ &= \frac{1}{N} \sum_{n=1}^N \left[\mathbb{I}_{(y_n=1)} \ln \left[\frac{1}{h(\mathbf{x}_n)} \right] + \mathbb{I}_{(y_n=-1)} \ln \left[\frac{1}{1 - h(\mathbf{x}_n)} \right] \right]. \end{aligned}$$

2.1.2 Cross-entropy and logistic regression

First we note the following property of the sigmoid function:

$$\begin{aligned} \theta(s) &= \frac{1}{1 + \exp \{-s\}} \implies \\ 1 - \theta(s) &= \frac{\exp \{-s\}}{1 + \exp \{-s\}} = \frac{1}{1 + \exp \{s\}} = \theta(-s). \end{aligned}$$

Using this and $y_n \in \{-1, 1\}$ we simply rewrite eq. (1) from the assignment:

$$\begin{aligned}\mathcal{L}(\mathbf{w}) &= \frac{1}{N} \sum_{n=1}^N \ln [1 + \exp \{-y_n \mathbf{w}^\top \mathbf{x}\}] \\ &= -\frac{1}{N} \sum_{n=1}^N [\mathbb{I}_{(y_n=1)} (1 + \exp \{-\mathbf{w}^\top \mathbf{x}\}) + \mathbb{I}_{(y_n=-1)} (1 + \exp \{\mathbf{w}^\top \mathbf{x}\})] \\ &= \frac{1}{N} \sum_{n=1}^N \left[\mathbb{I}_{(y_n=1)} \ln \left[\frac{1}{h(\mathbf{x}_n)} \right] + \mathbb{I}_{(y_n=-1)} \ln \left[\frac{1}{1 - h(\mathbf{x}_n)} \right] \right].\end{aligned}$$

Yielding the wanted result.

2.2 Logistic regression loss gradient (24 points)

2.2.1 Gradient for -1 and 1

Firstly we define a useful function $l_n(\cdot) \equiv \ln (1 + \exp \{-y_n \mathbf{w}^\top \mathbf{x}_n\})$ as the likelihood contribution of observation n such that

$$\begin{aligned}\mathcal{L}(\mathbf{w}) &= \frac{1}{N} \sum_{n=1}^N \ln (1 + \exp \{-y_n \mathbf{w}^\top \mathbf{x}_n\}) \\ &= \frac{1}{N} \sum_{n=1}^N l_n(\mathbf{w}),\end{aligned}$$

By the summation rule of differentiation we have

$$\nabla \mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N \nabla l_n(\mathbf{w}), \quad (1)$$

and by calculating an arbitrary of these N terms yields

$$\begin{aligned}\nabla l_n(\mathbf{w}) &= \left[\frac{\partial \mathcal{L}}{\partial w_1} \quad \frac{\partial \mathcal{L}}{\partial w_2} \quad \cdots \quad \frac{\partial \mathcal{L}}{\partial w_d} \right]^\top \\ &= \begin{bmatrix} \frac{1}{1 + \exp \{-y_n \mathbf{w}^\top \mathbf{x}_n\}} \exp \{-y_n \mathbf{w}^\top \mathbf{x}_n\} (-y_n x_{n1}) \\ \frac{1}{1 + \exp \{-y_n \mathbf{w}^\top \mathbf{x}_n\}} \exp \{-y_n \mathbf{w}^\top \mathbf{x}_n\} (-y_n x_{n2}) \\ \vdots \\ \frac{1}{1 + \exp \{-y_n \mathbf{w}^\top \mathbf{x}_n\}} \exp \{-y_n \mathbf{w}^\top \mathbf{x}_n\} (-y_n x_{nd}) \end{bmatrix} \\ &= -y_n \frac{\exp \{-y_n \mathbf{w}^\top \mathbf{x}_n\}}{1 + \exp \{-y_n \mathbf{w}^\top \mathbf{x}_n\}} [x_{n1} \quad x_{n2} \quad \cdots \quad x_{nd}]^\top \\ &= -\frac{y_n \mathbf{x}_n}{1 + \exp \{y_n \mathbf{w}^\top \mathbf{x}_n\}},\end{aligned}$$

which we insert into (1) to obtain the wanted result

$$\begin{aligned}
\mathcal{L}(\mathbf{w}) &= -\frac{1}{N} \sum_{n=1}^N \frac{y_n \mathbf{x}_n}{1 + \exp \{y_n \mathbf{w}^\top \mathbf{x}_n\}} \\
&= \frac{1}{N} \sum_{n=1}^N -y_n \mathbf{x}_n \frac{1}{1 + \exp \{-(-y_n \mathbf{w}^\top \mathbf{x}_n)\}} \\
&= \frac{1}{N} \sum_{n=1}^N -y_n \mathbf{x}_n \theta(-y_n \mathbf{w}^\top \mathbf{x}_n).
\end{aligned}$$

2.2.2 Gradient for 0 and 1

The exact same procedure is completed for the new encoding of the probability function:

$$\begin{aligned}
\mathbb{P}(Y = y_n \mid X = \mathbf{x}_n) &= \begin{cases} \theta(\mathbf{w}^\top \mathbf{x}) & , \text{ if } y = 1 \\ \theta(-\mathbf{w}^\top \mathbf{x}) & , \text{ if } y = 0 \end{cases} \\
&= [\theta(\mathbf{w}^\top \mathbf{x})]^{y_n} [\theta(-\mathbf{w}^\top \mathbf{x})]^{1-y_n}.
\end{aligned}$$

That is, we again setup the corresponding negative logarithmic likelihood function

$$\begin{aligned}
\mathcal{L}(\mathbf{w}) &= -\frac{1}{N} \sum_{n=1}^N \ln [\mathbb{P}(Y = y_n \mid X = \mathbf{x}_n; \mathbf{w})] \\
&= -\frac{1}{N} \sum_{n=1}^N \{ \ln [\theta(\mathbf{w}^\top \mathbf{x})^{y_n} \theta(-\mathbf{w}^\top \mathbf{x})^{1-y_n}] \} \\
&= -\frac{1}{N} \sum_{n=1}^N \{ y_n \ln [\theta(\mathbf{w}^\top \mathbf{x})] + (1 - y_n) \ln [\theta(-\mathbf{w}^\top \mathbf{x})] \},
\end{aligned}$$

and calculate the gradient

$$\begin{aligned}
\nabla \mathcal{L}(\mathbf{w}) &= -\frac{1}{N} \sum_{n=1}^N \begin{bmatrix} y_n \frac{1}{1+\exp\{-\mathbf{w}^\top \mathbf{x}\}} \exp\{-\mathbf{w}^\top \mathbf{x}\}(-x_{n1}) + (1-y_n) \frac{1}{1+\exp\{\mathbf{w}^\top \mathbf{x}\}} \exp\{\mathbf{w}^\top \mathbf{x}\}x_{n1} \\ y_n \frac{1}{1+\exp\{-\mathbf{w}^\top \mathbf{x}\}} \exp\{-\mathbf{w}^\top \mathbf{x}\}(-x_{n2}) + (1-y_n) \frac{1}{1+\exp\{\mathbf{w}^\top \mathbf{x}\}} \exp\{\mathbf{w}^\top \mathbf{x}\}x_{n2} \\ \vdots \\ y_n \frac{1}{1+\exp\{-\mathbf{w}^\top \mathbf{x}\}} \exp\{-\mathbf{w}^\top \mathbf{x}\}(-x_{nd}) + (1-y_n) \frac{1}{1+\exp\{\mathbf{w}^\top \mathbf{x}\}} \exp\{\mathbf{w}^\top \mathbf{x}\}x_{nd} \end{bmatrix} \\
&= -\frac{1}{N} \sum_{n=1}^N \left[\frac{1}{1+\exp\{-\mathbf{w}^\top \mathbf{x}\}} \right] \begin{bmatrix} (-\exp\{-\mathbf{w}^\top \mathbf{x}\}y_n + (1-y_n))x_{n1} \\ (-\exp\{-\mathbf{w}^\top \mathbf{x}\}y_n + (1-y_n))x_{n2} \\ \vdots \\ (-\exp\{-\mathbf{w}^\top \mathbf{x}\}y_n + (1-y_n))x_{nd} \end{bmatrix} \\
&= -\frac{1}{N} \sum_{n=1}^N \left[\frac{1}{1+\exp\{-\mathbf{w}^\top \mathbf{x}\}} \right] (y_n(1-\exp\{-\mathbf{w}^\top \mathbf{x}\}) + 1) \mathbf{x}_n \\
&= -\frac{1}{N} \sum_{n=1}^N \left[y_n + \frac{1}{1+\exp\{-\mathbf{w}^\top \mathbf{x}\}} \right] \mathbf{x}_n \\
&= -\frac{1}{N} \sum_{n=1}^N [y_n + \theta(\mathbf{w}^\top \mathbf{x})] \mathbf{x}_n.
\end{aligned}$$

Done!

2.2.3 Influence of misclassified examples

2.3 Log-odds (10 points)

By direct calculation we get

$$\begin{aligned}
\mathbf{w}^\top \mathbf{x} + b &= \ln \left[\frac{\mathbb{P}(Y=1 \mid X=x)}{\mathbb{P}(Y=0 \mid X=x)} \right] \iff \\
-(\mathbf{w}^\top \mathbf{x} + b) &= \ln \left[\frac{\mathbb{P}(Y=0 \mid X=x)}{\mathbb{P}(Y=1 \mid X=x)} \right] \iff \\
\mathbb{P}(Y=1 \mid X=x) \exp\{-(\mathbf{w}^\top \mathbf{x} + b)\} &= \mathbb{P}(Y=0 \mid X=x) = (1 - \mathbb{P}(Y=1 \mid X=x)) \iff \\
1 &= \mathbb{P}(Y=1 \mid X=x) (1 + \exp\{-(\mathbf{w}^\top \mathbf{x} + b)\}) \iff \\
\mathbb{P}(Y=1 \mid X=x) &= \frac{1}{1 + \exp\{-(\mathbf{w}^\top \mathbf{x} + b)\}} = \sigma(\mathbf{w}^\top \mathbf{x} + b).
\end{aligned}$$

That is if $\mathbf{w}^\top \mathbf{x} + b$ encodes the relative probability on log scale of observing $Y=1$ relative to $Y=0$ given $X=x$, then the sigmoid function describes the probability of observing $Y=1$.

3 Sleep Well (34 points)

3.1 Data understanding and preprocessing (4 points)

I am using python 3 and the numpy package to calculate the shares of observations in each of the five labeling categories.¹ The result is displayed in tabel 4. Notably the labels 0 and 2 are much more prevalent in the data and we therefore need to admit for sufficiently complex classification models to partition the data appropriately.

Category	Share
0	0.521
1	0.096
2	0.253
3	0.047
4	0.084

Table 1: Category shares

Note: The table displays the shares of observations labeled as each category in the training data.

3.2 Classification (30 points)

3.2.1 Logistic regression

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import GridSearchCV
```

Sample type	Risk
In Sample	0.149
Out of Sample	0.099

Table 2: Logistic regression empirical risk

Note:

3.2.2 Random forest

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
```

¹For the rest of part 3 i will use sci-kit learn package for the implementation of the machine learning algorithms.

Sample type	Risk (50 trees)	Risk (100 trees)	Risk (200 trees)
In Sample	0.155	0.156	0.155
OOB error	0.157	0.158	0.157
Out of Sample	0.097	0.097	0.098

Table 3: Random forest empirical risk

Note:

3.2.3 Nearest neighbor

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import GridSearchCV
```

Sample type	Risk
In Sample	0.150
Out of Sample	0.096

Table 4: K -NN empirical risk

Note: